



IoT Design Best Practices

An Analytical Report Generated by the AI Research Council

Transform Airports — AI Research Council

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Abstract

Most airport IoT deployments are stranded within 5 years — not because the sensors fail, but because the surrounding decisions (network, data ownership, integration path, vendor lock) were made for a pilot and never re-made for scale. The future-proofing question is not "what's the best sensor" but "what's the architecture that survives three vendor changes, two network upgrades, and one CIO turnover." The report examines this claim across the following: The stranded-deployment failure mode: catalog 6–10 major airport IoT deployments that became orphaned or got ripped out, and identify the specific architectural decision that doomed each; Network layer durability: LoRaWAN vs. cellular IoT (LTE-M, NB-IoT) vs. private 5G vs. Wi-Fi 6E for airport sensor applications. What survives a 10-year horizon and what's already obsolescing?; Data ownership and schema portability: how do airports that successfully migrate between platforms structure their data layer? Where does proprietary schema become a permanent tax?; The integration question: enterprise iPaaS vs. lightweight event brokers vs. direct database writes. What pattern do high-performing airport IoT programs actually use, and why does it differ from what enterprise IT recommends?; Edge compute placement: when does processing belong on-sensor, on-gateway, on-prem, or in cloud? Decision framework, not preferences.

How This Report Was Produced

This report is the product of the Transform Airports AI Research Council — a multi-agent analytical system running on Anthropic's Claude models. Rather than a single model writing in one pass, the Council convenes a panel of specialized research agents, subjects their work to adversarial review, and verifies every claim before publication. The process runs in four stages, with human review at two checkpoints.

Stage 1 — Independent research. Each research agent below produced an evidence brief without seeing the others' work, so the analysis draws on genuinely independent lines of evidence rather than a single line of reasoning.

Stage 2 — Synthesis and adversarial review. A Strategist synthesized the briefs into an argument; a Red Team attacked it across two rounds; the Strategist revised after each.

Stage 3 — Edit and fact-check. An Editor tightened the prose. A Fact-checker verified every number against the research briefs, removing or flagging what it could not source.

Stage 4 — Production. The verified draft was formatted into this document.

Research agents seated for this report

- Airline Commercial Strategist
- Airport CEO
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Process agents (on every report)

- **Strategist** — synthesized the research briefs into a single argumentative draft.
- **Red Team** — attacked each draft to surface weak claims and logical gaps.
- **Editor** — tightened the prose and removed filler without adding new content.
- **Fact-checker** — verified every numerical and attributed claim against the briefs.

Each agent is governed by a versioned specification — its analytical mandate, evidentiary standards, source discipline, and argumentative posture. The complete technical definitions for every agent on the Council are published and openly available at:

<https://github.com/TransformAirports/ai-council/tree/main/.claude/agents>

The Sensor Outlives the System

Transform Airports AI Council — IoT Design Best Practices June 2026

On February 22, 2022, AT&T switched off its 3G network. Verizon followed at year's end. No public record says what that did to any one airport's sensor inventory — and that absence is the first piece of evidence.

So picture it, faithful to the mechanism if not to a specific invoice. Somewhere in the asset-tracking and facilities telemetry of a large American hub, a population of devices installed before 2019 simply stopped reporting. The sensors had not failed. The batteries were fine. The mounting brackets held. What vanished was the radio underneath them — a decision made by a carrier in a boardroom no airport sat in. And because most 3G IoT devices cannot be rescued by swapping a SIM, the fix was not a software push. It was a technician on a ladder at every sensor location, one at a time, at airport labor rates, inside a Security Identification Display Area. No airport has published what that cost. The most likely reason is that the number was embarrassing.

That is the shape of the problem. Not the sensor. The architecture around the sensor.

The thesis is narrow on purpose. Among airport IoT programs that survive the pilot and actually scale, the dominant way they die is not hardware failure. It is the surrounding decisions — which network carries the data, who owns the schema it is written in, how it integrates with everything else, which vendor holds the keys, and who inside the airport is personally accountable for keeping it alive. Each gets made for a twelve-month proof of concept and never re-made for a twenty-five-year bond. The question that matters is not *what is the best sensor*. It is *what architecture — technical and organizational — survives multiple vendor changes, at least one network upgrade, and the departure of the people who built it*. An airport that cannot answer has not bought an operational-intelligence program. It has bought a future write-down with a warranty.

One honesty up front, because the whole report turns on it. How often scaled programs strand is not a number anyone can give you. Airports do not issue press releases when they rip out a sensor network; the failures leave the books quietly, folded into larger capital lines. The case that follows rests on named deployments and a traceable mechanism, not on a stranding rate. The limitation is real, and it cuts toward the skeptical reader, not away from him.

This is not a new failure. The industry has produced a fresh stranded generation in every technology wave for forty years. What is new is the surface area. The next failure will not strand the lighting data in one terminal. It will strand the gates, the baggage, the security queues, the ground fleet, and the digital twin — all at once, all on the same orphaned schema.

Executive Summary

1. The dominant failure mode in airport IoT is not sensor failure — it is platform discontinuation, protocol obsolescence, and schema lock, arriving on a schedule the vendor never put in the sales deck. Three extinction events landed inside eighteen months: Cisco wound down its Kinetic IoT platform (final order date May 2021) after exiting the smart-cities market in 2020; Sigfox, a low-power network operator with roughly ten million devices across seventy-plus countries, filed for bankruptcy in January 2022; and the US carriers completed their 3G shutdown by December 2022. No airport made any of those three decisions. Each stranded equipment that was working perfectly the day before.

2. The pilot-to-scale gap is the industry's modal outcome, but the stranding this report addresses happens *after* the pilot survives — and its rate is unobservable. Cisco's 2017 survey of 1,845 decision-makers found 60% of IoT projects stall at proof of concept and only 26% succeed completely; McKinsey's 2018 survey put the share stuck in pilot at 84%. Those measure programs that never scale. This report is about the smaller, costlier population that does scale, gets bonded, and then strands because the architecture was set in the pilot and never revisited — a population public data cannot size, because airports do not publicize rip-outs, so the case rests on the named catalog below.

3. The decision that strands a scaled program is not a technology choice — it is an authorization gap. When a pilot scales past the operating-budget threshold and below the board-approval threshold, it quietly becomes a capital liability amortized over 25–30 years against the debt-service-coverage ratios the rating agencies score. At a residual-cost airport, that liability flows straight to signatory-airline rates through majority-in-interest clauses. The Denver Great Hall P3 — terminated at 30% design completion, settled for \$183.6 million — is the governance failure in pure form: a long-horizon financial commitment made before the design was defined.

4. The hidden cost structure, not the license fee, bankrupts the program. Platform license fees are only 20–40% of three-year industrial-IoT total cost of ownership; the rest hides in integration, change management, and the engineering time to maintain custom connectors. Forrester puts IoT cost underestimation at 40–60%. A pilot-priced 1,000-sensor deployment runs 2.5–4× its Year-0 capital over ten years — and no airport feasibility study reviewed by the Chief Engineer lens modeled that ten-year figure. Toronto Pearson is the bill made visible: seven SCADA systems on three platforms accreted over fifteen years, consolidated only after converting 60,000 data tags and running parallel systems for twelve months.

5. The network layer is the most consequential decision, the one most often made implicitly, and the one most exposed to a politics the airport cannot influence. A LoRaWAN gateway needs power, backhaul, and a mounting location above a live baggage conveyor; private 5G needs licensed spectrum and equipment rooms; Wi-Fi 6E is for passenger devices, not sensors. Private LoRaWAN and 3GPP-family cellular (LTE-M, NB-IoT) survive a ten-year horizon; private 5G is correct only for

the high-bandwidth, low-latency workloads that justify it — and 75% of US private 5G rides on CBRS spectrum that the 2025 auction pipeline has put in play, the same federal hand that blocked C-band near airports for eighteen months. Durability is bounded from below too: a proprietary protocol on a fifteen-year PLC cannot be cleanly abstracted, so "survives two network upgrades" is a design target, not a baseline.

6. Data ownership is the least-discussed and most expensive dimension, and at airports it is legally unowned by default. Airline use-and-lease agreements contain no standard data-rights language, and IATA's October 2024 Aircraft Operational Data principles assert that data generated by an aircraft — in flight or on the ground — belongs to the airline. An airport that deploys sensors in a leased gate area, aggregates the data, and builds a platform on top is building on property it does not clearly own. ACRP's own research finds airport IoT data streams "frequently fall under vendor control," making airport access to its own data "challenging or costly."

7. There is a stranding mode unique to airports: the carrier already owns a better copy of the data. When an airport deploys sensors to measure what its dominant carrier already measures with higher fidelity — Delta's in-house Baggage AI runs a 99.9% scan rate on 120 million-plus annual bags; United migrated hundreds of applications to AWS for roughly \$2 billion in savings — the airport's parallel system never becomes the system of record. It produces data nobody uses. Not abandonment through distrust, not accretion through neglect: stranding by redundancy. No other industry has a dominant tenant who owns a superior parallel dataset and arguably owns the data rights too.

8. The architecture that survives is the most deliberately decoupled *and* the most clearly owned — and "open" is not free. Schiphol decoupled at the two seams that matter, the data schema and the transport broker, while buying *managed* simplicity above them; the Port of Rotterdam survives four simultaneous vendors because it declared the data layer to be port infrastructure; DFW's digital twin survives a CMMS upgrade only because its data layer is decoupled from the source systems. Five layers of vendor lock — hardware, connectivity, platform, schema, application — inflate five-year TCO by an estimated 20–30%, and the open, API-rich architecture that fixes lock also widens the attack surface the Seattle-Tacoma ransomware crew walked through in August 2024. The durable move: decouple where it counts, keep everything else simple enough for the team you actually have, write the exit cost into the contract before signing, and name one executive accountable for continuity across the leadership turnover that is a certainty.

The Argument

Start with the graveyard, because it is real, it is recent, and most of it is at airports

The most persuasive evidence for this thesis is not a forecast. It is a casualty list. Three cases are worth telling slowly, because the slow ones carry the mechanism. The rest are variations.

Begin at Toronto Pearson, because nothing failed there and everything stranded anyway. Over fifteen years of incremental baggage-system upgrades, each procurement team did the competent thing: it chose the best available SCADA platform for the project in front of it. No single choice was wrong. But not one procurement carried a sentence requiring the new system to share a supervisory interface with the last. So the systems accreted — seven independent SCADA platforms from three vendors by 2019, a layered estate Inductive Automation's case record calls "very hard and expensive to support, upgrade and maintain," carrying chronic latency. When Pearson moved to modernize under its Baggage 2025 program, it could not start on the baggage system at all. It had to first consolidate its own supervisory layer: 60,000 data tags converted, more than a hundred graphical displays translated, twelve months of parallel operation before the real work could begin. What doomed Pearson was not a product. It was an omission, repeated fifteen times — stranding by accretion, the quietest and most expensive way to arrive at a system no one can move.

Now Paris Orly, on a morning in November 2015, when a runway closed because of Windows 3.1. DECOR — the system linking Orly's air-traffic control to Météo-France for low-visibility operations — ran on an operating system Microsoft had abandoned two decades earlier. It failed, and a runway at one of Europe's busiest airports went dark until it could be coaxed back. The remediation was slow, and the reason is worth sitting with. France had three specialists who could work with DECOR. Three. One was retiring, with no identified successor. The knowledge of how the system worked lived not in documentation or a standards-based design any competent engineer could interrogate, but in three people's heads. When continuity depends on people instead of architecture, the system does not strand when the hardware dies. It strands when a person retires — and people always retire. This is what "survives one CIO turnover" undersells, because the person who actually holds the system is rarely the CIO. It is the network engineer who never wrote anything down.

The third slow case is Sigfox, the network that went bankrupt underneath its own customers. It raised more than \$300 million, built dedicated low-power networks in seventy-plus countries, and signed European airports and transport operators for perimeter monitoring and equipment telemetry. In January 2022 it filed for bankruptcy — a €91 million net loss on €24 million of revenue — was acquired for €25 million, and was back in court protection by 2025. The sin was singular: Sigfox used a non-standard air interface, so there was no fallback carrier. When the network wobbled, an airport could only keep depending on whoever owned the wreckage or replace the hardware. A

LoRaWAN deployment in the same spot swaps a network server. A Sigfox deployment swaps every device. The doomed decision was coupling to a proprietary radio with no second source.

The remaining cases are the same disease in different organs. Cisco shows mortality is not a startup problem: it exited the Kinetic for Cities market in 2020, wound down the broader Kinetic platform with a final order date in May 2021, extended support through 2024, and offered customers the chance to "transfer their data to a Cisco partner." That is the polite form of stranding — the vendor honors the contract and exits the market anyway, and no clause was breached because none was written for it. The 3G sunset, already told, is the radio treated as permanent infrastructure when it was a depreciating asset on a carrier's balance sheet.

Charlotte Douglas and Dallas Love Field are stranding-in-waiting. CLT is embedding 2,000 pavement sensors specified for a decade of life in a fourth parallel runway that will serve thirty to forty years, with no public answer to who owns the data pipeline after the university research program ends. Love Field runs the first airport private CBRS network, built and operated by Boingo — the most durable layer of the stack owned by a third party. And the original, Denver International's automated baggage system — a \$193 million BAE contract that scaled toward \$360–400 million, delayed the airport's opening by sixteen months, and was largely decommissioned within a decade — is the whole pathology at once: scope expanded mid-development from one concourse to three, integration treated as a residual, 72 certification tasks still incomplete at the expected commissioning date, no graceful way out because leaving cost more than staying. The bond obligations did not retire when the system did.

Then the case no other industry has. An undisclosed major hub deployed more than 400 Wi-Fi access points, 130 people-counters, and 50 Bluetooth sensors across its terminals to measure passenger flow — the iFlow system. (The deployment site is not named in the public record; iFlow's accuracy limitations were documented in a peer-reviewed study at a separate US airport.) The architecture worked. The sensors reported. But that study found the output lacked precision, the operations team quietly stopped trusting it, the vendor relationship atrophied, and the deployment now runs into a void where its users used to be. That is stranding by abandonment, and it teaches the lesson the counter-case will press: a system can have adequate architecture and still strand, because nobody trusted what it said.

But there is a fourth mode, the most distinctively airport failure of all. Install sensors in a gate area to track baggage the carrier already tracks better, and watch what happens. Delta's in-house Baggage AI runs a 99.9% scan rate on more than 120 million bags a year; United runs its operations on AWS infrastructure it migrated for roughly \$2 billion in savings. When the airport's sensor and the carrier's sensor record the same physical event, the carrier's record wins — higher fidelity, regulatory obligation, the customer relationship, the downstream integration. The airport's parallel deployment is an orphan not because it failed and not because anyone abandoned it, but because it was redundant the day it powered on. The cleanest documented version is AirAsia at Kuala Lumpur's

klia2, where a forced common-use transition collapsed self-service kiosk use from 35% to 10% and drove self-tagging to zero, because the carrier's own automation could not run inside the shared platform. Stranding by redundancy is the mode the technical lenses never see, because it is not a technology failure. It is a commercial one.

One disease, many organs. In not one case did the airport choose a *bad* sensor; in every one it coupled to something — a vendor, a protocol, a radio, a single person's memory, or a carrier that already owned the better dataset — at a layer it could not later decouple from without ripping out hardware or starting a fight.

History removes the novelty. After deregulation gutted Braniff, Eastern, Pan Am, and Midway in the 1980s, airports discovered that airline-specific check-in terminals were not transferable and wrote them off carrier by carrier. The answer was the common-use standard — CUTE, deployed at LAX for the 1984 Olympics: standardized hardware, airline logic on airline servers. The architecture was correct. But the standard took a decade to reach critical mass and required two more re-platformings — CUSS for kiosks, CUPPS in 2009 — each leaving its own stranded generation. The lesson — terminal technology must be vendor-agnostic by architecture, not by vendor promise — was learned, encoded in a standard, and then selectively forgotten with every new wave. The IoT question, MQTT or CoAP or proprietary transport, is CUTE's question one layer down.

And the original stranded IoT is still running. The building-automation systems installed in the 1990s terminal-construction wave, on LonTalk and Modbus and early BACnet, physically operate in every legacy terminal in the country, transmitting on protocols that cannot reach a modern IP platform without gateway hardware already a generation behind. Today's pilot is tomorrow's LonTalk: a sensor still faithfully reporting into a void where its platform used to be.

The decision that strands a scaled program is a bond decision wearing a technology costume

Here is the part the architecture conversation skips, because it happens above the architects' heads. When a pilot succeeds and scales, it crosses a threshold. Below it, the program was an operating expense — a few sensors a department bought on a discretionary line. Above it, it is capital, and capital at an airport joins the amortization schedule Moody's, S&P, and Fitch scrutinize, scored against a debt-service-coverage ratio large hubs target near 2.0x. A capital program that reliably produces five-year stranded assets while carrying 25-to-30-year bonds is a story no airport CEO wants to tell in a rating-agency briefing.

The trap is in the seam. A pilot that scales past the operating-budget threshold but below the board-approval threshold becomes a permanent financial commitment no one formally authorized. It succeeded, so it grew; it grew, so it got bonded; nobody re-asked the architecture question at the moment it stopped being an experiment. At a residual-cost airport, the liability flows to signatory carriers through the rate base — exactly what majority-in-interest clauses were built to police. An airport that arrives at an airline

partners committee with a \$75 million IoT remediation program, because the prior architecture required rip-and-replace, is negotiating from a posture it did not choose.

The Denver Great Hall is the governance failure stripped of its technology. Denver signed a \$1.8 billion concession with \$650 million of fixed construction at 30% design completion; eleven months in, the contractor asked for a \$288 million change order; the airport terminated and settled for \$183.6 million. The root cause its own CEO named — committing a long-horizon financial structure to an incompletely defined program — maps onto IoT exactly. CVG sharpens the credit point from the other side: the airport fell from A1 to Baa1 after Delta de-hubbed in 2013, having done nothing operationally wrong. One dominant counterparty reshaped the credit profile, and a network provider's insolvency does the same to every device on its network. Single-dependency risk in architecture is the credit analog of single-carrier concentration, and the rating agencies price both.

The personnel axis makes it worse than the slogan admits. CIO tenure at large US organizations averages under five years; for state-government CIOs the figure is about twenty months, with nearly 40% serving two years or less. A five-year IoT contract will see roughly two CIO cycles before its first renewal, and the incoming CIO has no incentive to inherit a predecessor's undocumented platform gracefully; the rational move is to declare it legacy and start fresh. So the survival standard is not "one CIO turnover." Plan for two. Each re-architecture appears in the capital program, gets bonded, and sits on the balance sheet for two decades. The technology officers will argue LoRaWAN versus 5G. The question upstream of all of them is who authorized the pilot architecture to become a permanent financial commitment, and at what threshold it stopped being a procurement decision and became a bond decision.

The cost is mostly invisible, which is why airports keep walking into it

The capital request hides the cost. The line items that win board approval are the visible ones — sensors, gateways, a platform license — the small fraction. License fees are only 20–40% of three-year industrial-IoT total cost of ownership. The other 60–80% hides in integration with the CMMS, FIDS, BMS, and airline systems; in change management through staff turnover; in retraining every time a vendor revs its API; and in the engineering hours to keep custom connectors alive. Forrester pegs the underestimation at 40–60%. A standard benchmark puts a 50-asset enterprise deployment at \$700K–\$1M over three years and a 1,000-asset airport at \$3–10M.

The ten-year version is worse, because it includes what a feasibility study omits: battery replacement at labor-dominant rates inside the SIDA; a gateway refresh at year five to seven driven not by hardware death but by protocol end-of-life; a platform renewal that arrives, predictably, at a 20–40% price increase once the airport is locked to the schema (the documented logistics case saw a 35% per-device jump at renewal, the vendor holding the device certificates); and firmware end-of-life that turns a working sensor into a security liability before the next capital cycle authorizes its replacement. Total ten-year cost of a pilot-priced deployment: 2.5 to 4 times Year-0. The simplest finding in the research set is the most telling: no airport feasibility study reviewed by the Chief

Engineer lens included a ten-year TCO model. Every one showed Year-0 capital plus a vague "O&M."

This is pilot purgatory stated in dollars. The pilot works because the systems integrator is still on-site and the gateway is hand-managed. The second terminal surfaces the real cost structure. Then the rational move — for both the airport and the vendor — is to renew rather than migrate, because re-exporting eighteen months of proprietary-schema sensor data costs more than signing again. Not a conspiracy; rational behavior by both sides when the exit cost was never made explicit at procurement. Toronto Pearson is the same lesson at enterprise scale: its prior structure of 150 separate IT vendors was so expensive to manage that consolidating to a single managed-services provider cut major outages 94%. The lesson is not that consolidation is always right — it is that fragmentation cost can grow severe enough to justify a very large check, and the check always comes due later than the pilot's economics implied.

The network decision is concrete, knowable, and usually made by accident

If one decision deserves the most engineering attention, it is the network — because it sets vendor optionality for a decade, and because it is physical infrastructure pretending to be a software choice. Every LoRaWAN-versus-5G debate eventually becomes a conversation about conduit, power feeds, and where you can mount a gateway above an active conveyor at three in the morning inside a secure zone. A major hub's communications plant is already fragmented across Wi-Fi, cellular DAS, proprietary OT networks, and hardwired control before the first sensor is placed; the 40%-plus sensor dropout rates reported in dense airport deployments are not discovered in production but designed in during procurement — and discovered at 5:45 on a Monday when the ops center starts getting alerts.

For once, the durability question has defensible answers rather than preferences. Private LoRaWAN is the strongest choice for high-density, low-bandwidth, facility-owned sensing: unlicensed spectrum, five-to-ten-year battery life, no recurring per-device fee, and an open LoRaWAN Alliance specification that lets an airport swap gateways or network servers without touching the sensor layer. LTE-M and NB-IoT now sit formally inside the 3GPP 5G family, tying their lifecycle to cellular evolution rather than one vendor's solvency; the cost trade is recurring carrier OpEx — manageable at a thousand sensors (roughly \$10–12 per device-year), a six-figure annual bill at ten thousand, the point where private LoRaWAN wins even after gateway capital. Private 5G is genuinely right for the workloads that need it: video analytics, autonomous apron vehicles, mobile assets demanding sub-five-millisecond latency and deterministic handoffs. Deploying it because it is "future-proof" and then using it to track restroom occupancy is capital misallocation. Wi-Fi 6E is for passenger devices; treating it as a sensor network inherits security exposure and the regulatory fragility iOS 14's MAC randomization demonstrated when it broke triangulation-based people-counting overnight.

Three honest qualifications, because the network decision is where the thesis is most often oversold. First, the sunsets are not hypothetical and they do not spare the "safe"

choices: GSMA Intelligence counts well over a hundred global 2G/3G shutdowns scheduled through 2030, and some carriers began retiring even LTE-M and NB-IoT in 2023–2024. Every connectivity standard has a finite life; the architecture has to survive the transition, which means decoupling the sensor layer from the transport layer with a standardized broker that outlives the radio beneath it.

Second — the qualification the cost comparisons never make — the most dangerous network risk at an airport is not technical obsolescence. It is spectrum politics, and the airport has no seat at the table. Seventy-five percent of US private 5G rides on CBRS, and the Trump administration's 800 MHz auction pipeline, written into the 2025 spectrum legislation, has — in the Wireless Internet Service Providers Association's own warning — "potentially put CBRS and 6 GHz spectrum on the auction block." The precedent is not theoretical. From January 2022 to September 2023, the FAA unilaterally blocked C-band 5G near airports over radio-altimeter interference, with no prior notice to airports or carriers, and the dispute resolved only after airlines spent eighteen months retrofitting altimeters. The FCC can auction spectrum; the FAA can block its use near airports; neither coordinates to protect an airport's capital plan; and the airport has no standing to object to either. An airport that engineers its entire sensor coverage map to CBRS power levels has staked a fifteen-year asset on a license that is a political decision, not a legal permanence. This is the real exposure underneath Dallas Love Field: the operating-agreement risk with Boingo sits on top of a spectrum-allocation risk neither Boingo nor the airport controls.

Third, durability is bounded from below. If the underlying hardware speaks a proprietary protocol — Modbus RTU on a fifteen-year PLC, a BACnet profile only one manufacturer's gateway can parse — the abstraction layer sits on something that cannot be cleanly abstracted. For most installations the realistic standard is one vendor change clean and one network upgrade with moderate pain. "Two clean network upgrades" is a design target to write into specifications, not a baseline to promise a board.

The standards are converging — which is a reason to write to them now, not to wait

There is a respectable argument, raised inside the research set, that the right move is restraint: delay major IoT commitments eighteen to thirty-six months while the standards consolidate, pilot only in brownfield, and write optionality into anything signed today. It has a real event behind it. The OPC Foundation completed protocol mappings for both LoRaWAN and Matter in 2025–2026, so the UA Edge Translator now bridges LoRaWAN, cellular, and Matter through OPC-UA — the industrial-IoT analog of FHIR becoming the interoperability layer for electronic medical records. If OPC-UA becomes the de facto integration layer, the architecture question collapses to one line: are you building to OPC-UA or not?

The cautionary tale it carries is exact. Hospital CIOs made twenty-year EMR commitments in the years just before the federal interoperability mandate reshaped the market, and many bought into the wrong layer at the wrong moment. An airport pouring concrete today could repeat the mistake.

But restraint fails the test of where airports actually are. The slab is being poured now. An airport mid-program cannot suspend its capital schedule for three years while a standards body finishes, and the conduit, equipment rooms, and network backbone going in today will be there in 2040 regardless of when the standards settle. The honest reconciliation is not "wait." It is "build to the converging standard." Write OPC-UA, ACRIS, and SWIM compatibility into the specification, keep the schema portable, and put the optionality in the contract — and the convergence stops being a reason to delay and becomes the reason to insist on open interfaces in the procurement running right now.

The seam that decides everything: the data schema and the integration broker

Here is where durable programs diverge from the rest, and the divergence is narrower than the marketing implies. It is not "buy a platform." It is two deliberate choices about where data lives and how it moves.

First, the schema. When an airport writes its operational history into a proprietary data model, the switching cost compounds every month, because migration is never just buying the new platform — it is running the old one in parallel, building translation layers to prove fidelity, and re-integrating every downstream consumer, several of them FAA-regulated systems carrying their own change-management overhead. The conservative rule of thumb: budget twice the vendor's integration quote. The remedy is not exotic, and the proof comes from the hardest possible environment. A legacy proprietary airfield-lighting control system, retrofitted with a gateway translating its closed protocol into standard AMQP and REST/JSON, hit end-to-end latencies of 43 milliseconds — inside the 500-millisecond regulatory limit for stop-bar operations — while preserving the legacy hardware and creating a portable data surface, at a migration cost below the cost of continued lock-in. That gateway is the whole thesis on one bench: not a rip-and-replace, not a maximal-optionality cathedral, just a closed protocol turned portable at a seam, cheaper than staying locked.

Second, the broker. Enterprise IT will recommend its familiar tools — MuleSoft, Boomi, Informatica — built for synchronous, moderate-volume API integration between enterprise systems. IoT is asynchronous, high-volume, and latency-sensitive; forcing it through a synchronous iPaaS adds latency, creates a single point of failure, and produces integration contracts that break across firmware releases. The pattern durable programs actually use is the opposite: MQTT at the edge, a lightweight event broker (MQTT or Kafka) for the real-time stream, and downstream consumption by SCADA, BMS, AODB, and analytics. The industrial-IoT community calls the mature version the Unified Namespace — an event-driven publish/subscribe topic hierarchy serving as a single source of truth, a design pattern, not a product anyone sells. The load-bearing principle is one sentence: the event broker is infrastructure, not an application, and it must not be owned by the vendor that owns the sensors or the analytics. When one vendor supplies sensor, integration, and dashboard, the airport has consolidated three dependencies into one contract, and one acquisition strands all three at once.

Here the most dangerous internal contradiction has to be faced, because the recommended exemplar appears to violate the recommended rule. Durable programs do not run their own broker on bare metal — Schiphol explicitly migrated away from self-hosted Apache Kafka to managed Confluent Cloud, for cost, elasticity, and staffing reality. A managed cloud service is precisely the risk class that killed Google Cloud IoT Core. So why is one acceptable and the other not? Google IoT Core coupled its customers through proprietary device SDKs embedded deep in firmware, so exit meant touching every device. Schiphol kept the Kafka schema open and the data continuously exportable, so the broker is disposable — if Confluent reprices, gets acquired, or shuts down, the schema moves to AWS MSK or Azure Event Hubs or self-hosted Kafka and the sensors never know. The rule: a managed broker can be disposable *if and only if* the schema is genuinely portable and the data continuously exportable. Not "avoid managed services." "Never let a managed service own your schema."

Edge compute: the decision framework, not a preference

The run asked for a framework, not opinions, and it is clean enough to put on one slide and defend in a design review. One rule governs it: processing that must survive a WAN outage belongs at the edge. Everything else follows from where each function sits relative to that line.

- **On-sensor.** Anomaly detection on continuous signals — vibration, temperature thresholds, acoustic signatures. The justification is bandwidth as much as latency: on-sensor inference cuts data egress by more than 90% on high-frequency signals, the difference between a battery that lasts five years and one that lasts eighteen months. Put here anything where the raw signal is high-volume and only the exception matters.
- **On-gateway.** Aggregation, protocol translation, and time-sensitive classification — passenger queue depth, equipment-fault categorization, the AMQP/REST translation that gives a legacy controller a portable data surface. This is the tier that buys decoupling: where a proprietary southbound protocol becomes a standard northbound stream, the layer that can be re-pointed at a new broker without touching the sensors.
- **On-premises.** SCADA and BMS control loops, real-time operational dashboards, the data retention regulators require. The test is binary: if the function must keep working when the internet does not, it lives here. Stop-bar lighting, HVAC setpoints, and access-control triggers cannot accept a cloud round-trip in their decision loop. An airport that puts operational control logic in a cloud-only deployment has designed for the nominal case and ignored the failure mode — and the failure mode is a Pacific Northwest airport's employees handwriting boarding passes during the Seattle-Tacoma ransomware attack.
- **Cloud.** Analytics, model training, multi-airport benchmarking, reporting. Everything with no operational dependency — and nothing with one. The cloud is where value compounds precisely because nothing in the running airport waits on it.

The framework also resolves a network argument the vendors muddy. A checkpoint with forty 4K camera feeds appears to demand fiber or 5G to move the video. Process it on-premises with edge GPUs and the network requirement collapses from hundreds of megabits of raw video to a few kilobits of metadata. The edge-placement decision and the network-sizing decision are the same decision, and getting the first right shrinks the second.

The two contract sentences that decide whether migration is even possible

Then the contract, because the architecture is downstream of it. Two sentences in the procurement decide whether the airport can ever leave, and both are usually missing.

The first is about hardware: *who owns the device certificates after termination?* When the vendor holds them, the airport cannot move its devices to a new platform without physically replacing hardware — the vendor holds every card, invisible until the day the airport tries to leave. The second is about data, and it needs teeth: not "does your platform support export" (every platform does) but "what is the liquidated-damages penalty if data cannot be exported in a documented, schema-portable format within thirty days of termination?" That question changes the vendor conversation, and it requires someone in the room who understands what "schema portability" means technically — exactly the person most airport procurement processes leave out. The SaaS world learned this a decade ago: bad architecture can be refactored; a bad contract runs for seven years. Open standards exist to make these clauses writable — ACRIS, SWIM, OPC-UA — and most RFPs reference none of them, because procurement officers do not know they exist and incumbent vendors have no incentive to volunteer the requirement that limits their own lock-in. Everything downstream — the schema seam, the broker, the migration path — is only as real as these two sentences make it.

The part the architects cannot solve: the architecture sits inside an organization

A decoupled architecture and a watertight contract will still strand if no human owns the system. This is the finding the technical lenses systematically miss, and three other industries learned it the hard way first.

Hospitals fought the ownership question for twenty years over their embedded medical devices — infusion pumps, monitors, imaging — which are IoT by any functional definition and outlive everything they talk to. IT said "it runs on our network, we own it." Clinical engineering said "it's clinical equipment, we own it." Both were right; the governance gap between them was where devices stranded. Roughly 30% of US hospitals resolved it by moving biomedical engineering under IT with a written governance agreement defining ownership, patching, and succession before any device went on the network. The other 70% are still fighting, and they hold the most stranded equipment. The airport parallel is exact: the gap between airport IT and the operational department that depends on the sensors — airfield ops, maintenance, terminal ops — is where airport IoT strands. Not because the sensors failed, but because nobody owned the gap, and the

ownership question surfaced in year three when something broke and both sides pointed at each other.

The Port of Rotterdam shows the resolved version. Its IoT program runs simultaneously on IBM, Cisco, Esri, and Axians, with the Port Authority explicitly owning the data layer as port infrastructure and operating a "live and let live" model: each stakeholder accesses only its relevant data through authorized APIs, and no single vendor controls the data model. That is not a technology architecture; it is a governance one. The program survives vendor failure because no single vendor is load-bearing — redundancy through diversification, a procurement posture, not an engineering trick. DFW proves the same principle inside a single program: its Willow/Parsons digital twin documented two non-negotiable requirements — an integration platform that survives external system replacements, and a data-conversion process spanning multiple spatial formats — and concluded that the \$2.9 million five-year program survives only if the data layer is decoupled from the source systems. When it isn't, a single CMMS upgrade orphans the entire twin. That is a US, scaled, operating proof point that the decoupling discipline is not a European luxury.

Which raises the objection this section has to answer rather than dodge. Every full-program governance exemplar above is a well-resourced public entity. Rotterdam is a semi-public corporation with the standing to declare "the data is infrastructure" and make it stick across four vendors. Schiphol bought managed Confluent because it has the staff to keep the schema open while outsourcing operations. Barcelona's open-source Sentilo platform — vendor-neutral, launched 2014, still running — took a city government with sustained civic will and a decade of commitment. A typical US mid-hub has none of that.

So the honest prescription cannot be "be Rotterdam." It is narrower and cheaper. For the airport without the team, the durable layer is not a platform it builds; it is a contract it signs and a person it names. The two contract sentences carry the optionality the missing engineers cannot. A single accountable owner — someone who can explain the system to an incoming CIO in thirty minutes, who has a defined successor, who owns the IT-operations gap in writing — carries the continuity the missing standing cannot. DFW and the airfield-lighting gateway prove the technical half is portable to US airports. The governance half is portable only if it is made small enough to fit the organization that actually exists.

This is also where the capital-handoff failure hides. A master-systems-integration analysis found that systems-integration requirements are "frequently omitted entirely" from airport program-definition documents, each contractor assuming someone else holds the integration safety net. ACRP Report 59 identified the pattern in 2011; the same analysis found it again in 2024. The thirteen-year gap is itself the finding: the industry invokes Denver's baggage system as a cautionary tale and then reproduces its preconditions in every major capital program, because the COO who will run the building arrives at commissioning rather than at program definition, when the integration requirement could still have been written down.

The synthesis the architects need to hear: the most durable airport IoT program is not the one with the cleverest protocol. It is the one a named executive fights to keep — someone who owns the gap between IT and operations in writing, signed, before the first sensor goes on the wall. People fight to keep what they own and understand. The architecture survives only inside an organization built to carry it.

The Counter-Case, Honestly Presented

The strongest argument against everything above is that it is a consultant's product — complexity sold by people whose revenue scales with the complexity they recommend — and that it misdiagnoses the disease. That case is not a strawman. In several places it is correct.

Begin with the word "stranded." Enterprise IT runs on refresh cycles by design: servers four to six years, network gear five to seven, laptops three to four. An IoT sensor network significantly upgraded at year five is not stranded; it is a planned refresh, the ordinary discipline applied to any asset. Define stranding as "eventually superseded" and every technology in history fails the test. The right question is not "did it survive five years?" but "did it return value during its operating life?" A building-management system that optimized HVAC loads for four years before a planned upgrade was not stranded. It was amortized.

Now turn the graveyard around, because it cuts the other way. BACnet was published in 1995, Modbus in 1979. Both still dominate airport building automation — decades of service in environments that clock ten to twenty years without overhaul. Against that, look at what the architecture-first thesis points airports toward: Google Cloud IoT Core, dead at six. IBM Watson IoT, sunset at roughly a decade with no direct replacement. Cisco Kinetic, retired. The single-vendor, "locked-in," proprietary building-management systems running quietly in airports worldwide have outlived every major cloud IoT platform of the last decade. The thesis tells airports to flee vendor lock-in into the arms of hyperscalers who have proved less durable than the systems they are fleeing.

The deeper objection is about cause. If stranding were primarily driven by wrong network choice or proprietary schema, programs would die in years three to five, when those decisions bite. Instead, most never reach production at all — Cisco's own data put successful completion at 26% and named unclear business goals, not integration architecture, as the leading cause. An airport that deploys queue sensors, builds a dashboard, and never connects the data to a gate-staffing decision is not suffering from vendor lock-in. It is suffering from unclear purpose, and a better integration stack would not have saved it. iFlow is the example the thesis itself has to concede: the data was technically available and operationally ignored. Fixing the architecture does not fix the purpose.

The thesis also assumes a cost structure that does not survive contact with the field. Open protocols, portable schemas, and multi-vendor stacks are not free. Every seam —

separate MQTT brokers, Kafka clusters, schema-translation layers, API gateways — adds overhead that scales with fleet size. Industry data finds IIoT projects underestimate total cost of ownership by 30–50%, and the gap widens in regulated environments. A single-vendor integrated deployment eliminates most of the seam cost: one logistics operator cut its IoT spend from roughly \$38,000 a month to \$8,000 by consolidating from a public-cloud service to a single on-premises solution. That is not, on its face, a lock-in risk. It can be a lock-in benefit.

And it assumes a workforce airports do not have. The prescription presumes in-house engineers who can write schemas, govern APIs, run brokers, and execute migrations. Almost no airport has them. The airport that builds a "vendor-change-surviving" architecture and then loses its one IoT architect to a tech company in year two has produced a stranded deployment of a different kind: an orphaned architecture too complex for the staff who remain. The admired benchmark cuts this way too. Chang's coordination platform runs thousands of sensors, hundreds of cameras, and hundreds of mobile devices as a deeply integrated single-platform deployment backed by a large internal technology organization — not a loosely coupled, multi-vendor mesh. The airports that do IoT well tend to commit deeply to one integrated platform and build capability around it, not to architect for an optionality they will almost never exercise.

The operator's bench adds three corrections the thesis must answer rather than dodge. First, **open architecture is its own attack surface**. The Seattle-Tacoma ransomware attack came through networked systems, not a closed proprietary one, and the federal systems that stayed up did so because they were segmented, not because they were open. Every open API the thesis recommends is a target; it cannot cite SEA-TAC as evidence for segmentation while ignoring that its own architecture widens the surface SEA-TAC exploited. Second, **"two network upgrades" is aspirational**. Where the hardware-layer protocol is proprietary, the abstraction layer sits on something that cannot be cleanly abstracted; the realistic version survives one vendor change cleanly and one network upgrade with moderate pain. Third, **data governance without organizational governance is worthless**. If an airport puts IoT data under the CIO, facilities under the COO, and airline-facing operational data under the VP of Airline Affairs, the contract's data-ownership provisions are irrelevant — nobody has unambiguous authority to act on unified data.

There is a fourth correction the airline lens adds, the most specific to airports: the airport may be building a system its dominant carrier will never feed or use. At a hub where one carrier controls 60%-plus of landed weight, the carrier already owns a superior parallel dataset and, under the IATA principles, arguably owns the rights to the gate-area data the sensors would capture. The history is unkind. Pittsburgh built \$1 billion of terminal to US Airways' hub specification in 1992; when the hub collapsed in 2004 the infrastructure had no second use, and the airline shed its maintenance obligations in bankruptcy, leaving the airport to maintain systems it had never owned. St. Louis went from 417 daily TWA flights to 36 and demolished a concourse. Infrastructure encoded to one carrier's logic strands when the logic changes — and a sensor network in a carrier's leased gates is exactly that kind of single-counterparty infrastructure.

The regulatory and political structure sharpens the same blade. Even an airport that wants the durable architecture must fight its own capital structure to fund it. The PFC cap has been frozen at \$4.50 since 2001 — to hold its original value against CPI it would need to be roughly \$7.85 in 2026 dollars — and the 2024 FAA reauthorization did not raise it. AIP eligibility funds runways, taxiways, lighting, conduit, and equipment rooms, not middleware, schema migration, or vendor-neutral integration layers. The IIJA Airport Terminal Program, the closest thing to federal co-investment in terminal technology, closed its final window on January 15, 2026. Rating agencies reward visible, bond-ratable revenue and view operational-technology spend skeptically unless it models clean OpEx reduction. The result is a structural bias: airports can fund the *hardware* of IoT and not the *durability* of it. Telling an airport to invest in integration-layer maintenance is telling it to fight uphill against its own bond covenants and a federal funding stack that just closed the one window that helped.

A reader who came in skeptical should feel the case has been made fairly. Now to answer it.

Why the Counter-Case Is Insufficient

Concede the real points, once and by name. The cloud graveyard is real; an airport that built a serious pipeline on Google Cloud IoT Core or IBM Watson IoT is stranded today. Most programs die before architecture matters, and the killer at that stage is unclear purpose, not protocol — iFlow proves a system can have architecture and still be abandoned. Open and multi-vendor stacks add real seam cost, and consolidation can genuinely cut spend. BACnet and Modbus have outlasted every cloud IoT platform. The physical layer comes first; staffing binds; open APIs widen the attack surface; "two network upgrades" is aspirational; the dominant carrier may own a better dataset and refuse to feed the airport's; organizational misalignment renders the best contract inert; the capital structure starves the durable layers. None of these is a quibble. The thesis survives all of them — because the counter-case has mostly refuted a claim the thesis does not make.

The thesis is not "choose open over proprietary," and it is not "buy the sophisticated multi-vendor platform." Read the counter-case's own conclusion: deploy against a clear outcome, staff for maintainability, and treat every hyperscaler managed IoT service as a platform risk. That last clause is the thesis. The two sides disagree about billing order more than substance — and the disagreement resolves once you see what the counter-case's best evidence actually shows.

Take the BACnet paradox, the counter-case's strongest shot. Why has Modbus survived since 1979 and BACnet since 1995 while Google IoT Core died at six? Not because they were proprietary and single-vendor, but because they are open, standardized, and decoupled from any one vendor's solvency — any manufacturer's device can speak them, and the protocol outlives them all. The durable thing in every one of the counter-case's own examples is the open, decoupled interface; the dead thing is the proprietary, vendor-

coupled cloud platform. The lesson the counter-case draws — embrace single-vendor lock-in — is the opposite of what its evidence shows. Durability comes from decoupling at the interface, not from choosing a vendor and praying it survives. The airfield-lighting gateway proves it on a bench — a closed protocol turned portable at 43 milliseconds, inside the regulatory limit, cheaper than staying locked. Not maximal optionality. One seam, decoupled, in the hardest environment there is.

Which reframes the disagreement. "Maximum optionality" really is a consultant's product — expensive, complex, unstaffable, rarely exercised. But it is a caricature of the thesis. The actual prescription is *minimal coupling at the two or three seams that matter, and ruthless simplicity everywhere else*. Decouple the data schema, because that is what compounds and what FAA-regulated downstream systems depend on. Decouple the transport broker from the sensor vendor, because that is what survives the radio sunset and the platform death. Keep operational control logic at the edge, because that is what survives the WAN outage. And then make everything above those seams as simple as the team you actually have can run. Schiphol is not a maximal-optionality cathedral; it bought *managed* Confluent precisely for the staffing reality while keeping the schema open. Durability is not complexity. Conflating the two is the flaw in an otherwise serious objection.

On "most programs die before architecture matters" — concede the number, reject the inference. The thesis is not addressed to the program that dies in the pilot. It is addressed to the one that *survives* the purpose gate, gets sponsorship, deploys, scales, and then strands because it made its architecture decisions back when it was a pilot and never re-made them. The counter-case's own list of scenarios where it admits the thesis is correct — cloud-lock with a platform that fails, network stranding on proprietary narrow-band, baggage-schema continuity across a vendor change, multi-terminal phased expansion — are not edge cases at a large hub. They are its operating conditions. You write an architecture standard not for the deployments that die in the crib but for the ones that live long enough to strand expensively.

The operator and airline corrections do not defeat the thesis; they finish it. On the attack surface: design decoupling and segmentation as one discipline — SEA-TAC is evidence that the OT/IT boundary must be engineered into the same design that opens the schema, which is why TSA's segmentation directive and the data-export clause belong in the same RFP. On "two network upgrades is aspirational": concede it, and write the slogan into specifications as "survive one vendor change cleanly, one network upgrade with manageable pain, second upgrade as an aspiration." On the carrier owning the better dataset: not a reason to skip the architecture but a reason to put the carrier-consent and data-rights question *ahead* of sensor selection, because a sensor that generates data the airport cannot use or legally own is stranded the day it ships. On organizational governance, the most serious correction: contract clauses are necessary and insufficient — they require a designated internal data owner and a master systems integrator with real authority, named before construction, plus board-level authorization discipline so the pilot cannot scale into a bonded liability without anyone deciding to let it. That is the half the technical lenses omit, and the thesis is stronger for absorbing it.

Which leaves the capital-structure objection — the one the thesis cannot argue away and should not try. The PFC freeze, the AIP exclusion, and the closed IIJA window are real, and they starve the durable layers. But notice what the argument establishes. It does not show the architecture is wrong; it shows it is *underfunded* — that airports systematically buy the sensor hardware (AIP-adjacent, bond-ratable) and skimp on the schema and integration layers (neither), and then strand. That is not a refutation of the thesis. It is the thesis's causal mechanism, confirmed by an independent lens. And the timing inverts the easy reading: because the federal terminal-technology window has closed, there is no grant to fund a re-do. An airport designing into that cliff has *more* reason to get the durable layers right the first time — through the operating budget, through the integration line of a capital project, through writing data-layer requirements into the RFP so the cost is bid rather than discovered. The constraint makes the discipline more necessary, not less.

So the thesis stands, narrowed and sharpened by the attack: not "build for maximum optionality," but decouple at the schema and the broker; keep everything above those seams simple enough to staff; engineer segmentation into the same design that opens the data; settle the carrier data-rights question before the sensor question; name a single accountable owner with a defined successor; make exit cost contractual; and fund the durable layers on purpose, because the system will not. The naive version deserved to lose. The narrowed version is stronger for having survived.

Implications for MWAA and Washington Dulles

This is not an abstract debate for the Metropolitan Washington Airports Authority. It is a decision being poured in concrete right now.

MWAA's new use-and-lease agreement took effect January 1, 2025, governing a \$9 billion capital program with roughly \$6.99 billion of it at Dulles. The centerpiece for the carrier that matters is United's \$500-million-plus Concourse E expansion: fourteen exclusive-use gates in a 435,000-square-foot facility, announced December 2024, defining the physical and operational envelope of Dulles for the next decade. It is the highest-stakes IoT architecture decision MWAA will make this cycle. The conduit runs, equipment rooms, structured wiring, and network backbone are all being specified today under FAA design standards — far cheaper to get right in design than to retrofit.

The historical rhyme is real, but the funding posture has flipped, and the flip strengthens the argument. The conditions that produced stranded explosive-detection installations in 2002 were capital abundance and compressed timelines. MWAA in 2026 faces the opposite: the IIJA Airport Terminal Program closed its final window on January 15, 2026, so the federal cost-share that might have funded Concourse E technology is gone, and a frozen PFC cannot make up the difference. MWAA is building into the funding cliff, not riding a wave of it. That is precisely why durable architecture is more urgent here — there will be no federal money for a re-do, so the data layer has to be right the first time, on bonded capital the rating agencies will score.

Three specific exposures follow.

First, the data-rights vacuum is acute at Dulles precisely because United is dominant. Those fourteen gates are United's leased premises, and IATA's October 2024 principles assign the airline ownership of data an aircraft generates on the ground — docking time, bridge engagement, ground-equipment positioning. United has its own AWS-based operational data infrastructure and roughly \$2 billion in booked technology savings; it does not need MWAA's. So any sensor MWAA installs in Concourse E is a parallel system generating duplicate data in a space where United controls the operational reality and arguably owns the data — stranding by redundancy at the most expensive possible scale. Pittsburgh is the warning: a billion dollars of terminal built to one carrier's hub logic, stranded when the logic changed, the airport left maintaining systems it never owned. Whether or not United alone clears the formal majority-in-interest threshold, the practical power points the same way, which makes Concourse E technology a substantially bilateral negotiation. MWAA should settle the United consent-and-data question in the lease language *before* deploying, not after a legal notice arrives. The architecture problem is downstream of the contract problem.

Second, fund the durable layers on purpose, because the capital structure will not. Concourse E's AIP-adjacent elements — conduit, equipment rooms, structured wiring — will get funded. The schema design, the integration broker, the vendor-neutral middleware, and the certificate-ownership and data-export clauses determine whether the deployment survives to 2040, and those are exactly what AIP and a frozen PFC do not cover. MWAA should write data-layer requirements into the Concourse E technology RFP so the integration cost is bid and visible rather than discovered at the second concourse: reference the ACRIS Semantic Model, FAA SWIM schemas, and OPC-UA explicitly, require open-schema export with liquidated damages, and require airport ownership of device certificates at termination. Writing to OPC-UA also answers the "should we wait for the standards to settle" question without waiting. MWAA can close the gap with a paragraph in the specification — paired with a named internal data owner and a master systems integrator engaged at program definition, not commissioning, because the clause is inert without someone holding authority to enforce it.

Third, the segmentation clock is already running. TSA's 2023 cybersecurity directive made IT/OT segmentation mandatory; a forthcoming FAA rulemaking, chartered under the 2024 reauthorization, will impose a second pass later this decade. A flat-network sensor deployment in Concourse E would be born out of compliance and face two rounds of forced remediation inside its first five years. Seattle-Tacoma's August 2024 ransomware attack — dry-erase boards, 90,000 breach notifications, manual baggage handling at a peer hub — shows what the seam costs when a threat actor finds it. The lesson cuts both ways: the open, API-rich data layer that keeps Concourse E from stranding is also the layer that widens the attack surface, so segmentation and schema-portability are one design problem, not a trade-off. The CISO is already a veto voter on this architecture whether the program plan acknowledges it or not; designing segmentation in now is cheaper than the "architecture review" pause where airport IoT goes to die quietly.

The concrete recommendation for MWAA is narrow and cheap relative to the \$6.99 billion it sits inside. On Concourse E, before the build locks: separate sensor procurement from platform procurement; require open-schema data export and airport ownership of device certificates at termination, with liquidated damages; specify ACRIS/SWIM/OPC-UA-compatible APIs; decouple the transport broker from the sensor vendor; avoid staking the sensor coverage map on CBRs power levels alone, given the auction risk; segment OT from IT to current TSA standard within that same design; name an accountable internal data owner with a defined successor; and settle the United data-rights question in writing. None of that is a moonshot. It is procurement and governance discipline, exercised once, at the moment MWAA has the most bargaining power it will ever have — before the contract is signed and the schema starts filling up.

The sensors going into Concourse E in the next two years will almost certainly still be bolted to the walls in 2040. The question is whether anyone will still be able to read what they are saying — or whether they will have joined the controllers in the old terminals and the system at Orly that three people understood, faithfully reporting, decade after decade, into a platform that no longer exists, in a language no one kept the key to.

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